



5558 - A Census of Optical Diagnostics of Ionizing Sources Across Cosmic Time

Cycle: 3, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Nikko J. Cleri (PI)	Texas A & M University
Dr. Anton M. Koekemoer (CoI)	Space Telescope Science Institute
Antonello Calabro' (CoI) (ESA Member)	INAF, Osservatorio Astronomico di Roma
Bren Backhaus (CoI)	University of Connecticut
Dr. Casey Papovich (CoI)	Texas A & M University
Prof. Dale D. Kocevski (CoI)	Colby College
Dr. Grace Olivier (CoI)	Texas A & M University
Dr. Denis Burgarella (CoI) (ESA Member)	Laboratoire d'Astrophysique de Marseille
Prof. Jonathan R Trump (CoI)	University of Connecticut
Kelcey Davis (CoI)	University of Connecticut
Lise-Marie Seille (CoI)	Laboratoire d'Astrophysique de Marseille (LAM)
Dr. Mark Dickinson (CoI)	NOIRLab - (AZ)
Dr. Michaela Hirschmann (CoI) (ESA Member)	Ecole Polytechnique Federale de Lausanne
Pablo Arrabal Haro (CoI)	NOIRLab - (AZ)
Dr. Raymond Simons (CoI)	University of Connecticut
Dr. Rebecca L. Larson (CoI)	Rochester Institute of Technology
Prof. Steven L. Finkelstein (CoI)	University of Texas at Austin
Dr. Taylor Alexandra Hutchison (CoI)	NASA Goddard Space Flight Center
Prof. Veronique Buat (CoI) (ESA Member)	Laboratoire d'Astrophysique de Marseille
Dr. Vital Fernandez (CoI)	University of Michigan Ann Arbor
Dr. Weida Hu (CoI)	Texas A & M University

OBSERVATIONS

ABSTRACT

We propose a systematic improvement to the selection of active galactic nuclei (AGN) throughout cosmic time via rest-frame optical emission-line ratio diagnostics. We will (1) perform photoionization modeling to describe the evolution of the BPT, VO87, and OHNO diagrams using physically motivated scenarios, including alpha- or CNO-enhanced chemical abundances of stellar populations and AGN, (2) leverage publicly available spectroscopy of $\sim 14,000$, $0 < z < 10$ galaxies from the JWST archive to test their evolution of BPT, VO87, and OHNO as a function of observable characteristics, and (3) combine these methodologies to formulate a comprehensive physics and chemistry informed redshift evolution of BPT, VO87, and OHNO. This work will provide a critical result to the community which is immediately applicable to all studies of high-redshift emission-line galaxies.

OBSERVING DESCRIPTION